

Schema Design

User

(user_id, first_name, last_name, email, password_hash)

Primary Key: user_id

Household

(household_id, user_id, region, housing_type, number_of_residents, primary_cooking_fuel, ac_type, water_heater_type, last_updated)

Primary Key: household_id

Foreign Key: user_id References User (user_id)

Bill

(bill_id, user_id, month, year, total_kwh, total_cost_sar, season, avg_temp, created_at)

Primary Key: bill_id

Foreign Key: user_id References User (user_id)

Insight

(insight_id, consumption_pattern, comparison_to_region, insight_text, bill_id, household_id)

Primary Key: insight_id

Foreign Key: bill_id References Bill (bill_id)

Foreign Key: household_id References Household (household_id)

Advice

(advice_id, advice_text, is_applied, applied_date, bill_id, household_id)

Primary Key: advice_id

Foreign Key: bill_id References Bill (bill_id)

Foreign Key: household_id References Household (household_id)

Visualization

(visual_id, chart_type, chart_data_json, image_path, bill_id, household_id)

Primary Key: visual_id

Foreign Key: bill_id References Bill (bill_id)

Foreign Key: household_id References Household (household_id)